

LANGEBAAWEG, South Africa

WMO: **687140**Lat: **32.9725S**Lon: **18.1578E**Elev: **31**StdP: **100.95**Time Zone: **2.00 (E02)**Period: **96-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	5.1	-0.6	3.6	20.0	0.8	4.0	15.9	11.0	15.3	10.0	15.4	1.3	190	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.2	35.0	19.2	32.8	18.7	30.8	18.1	20.9	30.5	20.1	28.8	19.5	27.4	5.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.1	22.3	17.6	12.6	21.9	17.0	12.1	21.5	60.2	30.7	57.8	29.1	55.7	27.6	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.6	8.6	2.1	39.5	1.1	1.4	1.3	40.5	0.6	41.3	-0.1	42.0	-0.9	43.0
			1.3	22.1	1.3	0.8	0.4	22.7	-0.4	23.2	-1.1	23.7	-2.0	24.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	21.8	22.1	20.7	18.6	15.6	13.2	12.7	12.9	14.7	17.0	18.5	20.7	(6)
		DBStd	4.34	2.59	2.73	2.93	3.38	2.68	2.22	2.51	2.31	2.92	3.23	2.65	2.56	
		HDD10.0	7	0	0	0	0	0	1	3	2	1	0	0	0	
		HDD18.3	851	2	1	7	36	93	155	174	169	116	64	28	6	
		CDD10.0	2697	366	339	333	259	175	97	88	92	142	218	256	331	
		CDD18.3	499	108	107	81	45	9	1	1	1	8	24	34	78	
		CDH23.3	4687	1001	937	700	406	114	24	23	36	109	293	367	676	
		CDH26.7	1777	406	391	270	159	28	2	5	35	109	126	245		
(14)	Wind	WSAvg	3.7	4.8	4.5	3.9	3.3	3.0	2.9	2.9	3.1	3.4	4.0	4.5	4.6	(14)
(15)	Precipitation	PrecAvg	267	7	5	8	18	34	49	45	25	12	12	7		(15)
(16)		PrecMax	417	41	28	22	55	78	103	117	87	74	53	38	23	(16)
(17)		PrecMin	117	0	0	0	3	2	6	12	9	4	0	1	0	(17)
(18)		PrecStd	72	9	7	7	12	19	22	23	19	19	12	10	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	37.6	37.3	36.8	34.6	29.9	26.2	26.0	27.1	30.7	34.1	33.9	35.4		(19)
(20)		0.4% MCWB	20.6	20.3	19.7	18.0	16.0	14.2	14.5	15.1	15.7	17.6	18.0	19.3		(20)
(21)		2% DB	33.7	34.1	32.7	31.1	26.8	23.2	23.1	23.3	26.2	30.1	30.2	32.1		(21)
(22)		2% MCWB	19.1	19.7	19.1	17.4	15.3	13.9	13.8	14.1	15.6	16.5	16.9	18.7		(22)
(23)		5% DB	31.1	31.3	29.5	28.0	23.7	20.1	20.7	20.1	23.1	26.9	27.6	29.4		(23)
(24)		5% MCWB	19.0	19.1	18.1	16.7	14.8	13.3	13.4	13.5	14.8	15.7	16.4	18.2		(24)
(25)		10% DB	28.6	28.8	27.1	25.2	20.9	17.8	17.9	17.8	20.2	23.9	25.0	27.0		(25)
(26)		10% MCWB	18.3	18.8	18.0	16.2	14.7	13.0	12.7	13.1	14.0	15.3	16.1	17.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	22.1	22.3	21.1	19.8	17.7	16.2	15.9	16.6	18.2	19.1	19.9	20.8		(27)
(28)		0.4% MCDB	32.5	31.9	32.4	29.3	24.7	21.6	22.2	23.5	26.8	30.4	28.4	31.1		(28)
(29)		2% WB	20.6	20.9	20.1	18.7	16.8	15.2	14.8	15.2	16.5	17.7	18.3	19.8		(29)
(30)		2% MCDB	29.9	30.1	28.8	27.2	23.0	19.1	19.8	20.2	23.4	26.2	25.9	28.6		(30)
(31)		5% WB	19.8	20.1	19.4	17.8	16.0	14.5	14.0	14.3	15.3	16.7	17.5	19.1		(31)
(32)		5% MCDB	27.9	28.3	26.9	24.8	21.0	17.7	18.4	18.3	21.0	23.9	24.4	26.6		(32)
(33)		10% WB	19.1	19.4	18.7	17.0	15.3	13.9	13.3	13.5	14.5	15.8	16.7	18.4		(33)
(34)		10% MCDB	26.1	26.8	25.2	22.8	19.1	16.6	16.9	16.9	19.1	21.8	22.8	25.1		(34)
(35)	Mean Daily Temperature Range	MDBR	12.2	12.2	12.0	11.7	10.4	10.1	10.8	10.3	11.2	11.9	11.5	11.7		(35)
(36)		5% DB MCDBR	16.6	16.3	16.0	16.2	15.6	15.1	15.5	15.5	16.2	17.2	16.6	16.1		(36)
(37)		5% DB MCWBR	5.5	5.3	5.4	5.7	6.9	7.8	8.1	8.2	7.3	6.7	6.1	5.5		(37)
(38)		5% WB MCDBR	13.3	13.3	13.0	13.2	12.1	10.1	12.2	12.1	13.2	13.7	12.9	12.9		(38)
(39)		5% WB MCWBR	5.1	5.1	5.1	5.6	6.5	6.4	7.5	7.3	6.9	6.1	5.4	4.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.336	0.331	0.318	0.315	0.309	0.312	0.329	0.338	0.348	0.332	0.336		(40)
(41)		taud	2.534	2.540	2.550	2.581	2.581	2.570	2.555	2.509	2.446	2.435	2.522	2.526		(41)
(42)		Ebn at Noon	1002	981	952	911	860	839	854	884	928	958	999	1005		(42)
(43)		Edh at Noon	110	106	98	85	75	72	76	90	107	116	111	112		(43)
(44)	All-Sky Solar Radiation	RadAvg	8.31	7.48	6.07	4.46	3.12	2.66	2.97	3.71	5.08	6.59	7.81	8.39		(44)
(45)		RadStd	0.21	0.20	0.20	0.23	0.21	0.15	0.17	0.17	0.19	0.29	0.22	0.26		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		+0.36	+0.49	N/A	N/A	+0.53	+0.47	-2	-74	+111	+61

Nomenclature: See separate page